COMS4995W31
Applied Machine Learning

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Columbia University | Spring 2026

Announcement

Assignment 2 is on the way…

Assignment 1 grades will be released in the next 2 days.

AI of the Week: The Hidden "Workhorse" of Modern AI

While the tech world obsesses over the "Attention" mechanisms in models like Transformers, the unsung heroes doing the heavy lifting are the fundamental structures we are learning today, namely

Multilayer Perceptrons and **Backpropagation**

Transformers are Powered by MLPs

- The majority of the parameter count actually resides in the feed-forward neural networks
- These stack simple neurons to transform raw text features into highly abstract representations

Non-Linearity is the Secret to "Understanding"

- Real-world data is highly complex and non-linear
- Industry models rely heavily on modern variants of the ReLU activation, such as GELU, to give the network its flexible decision surfaces

AI of the Week: The Hidden "Workhorse" of Modern AI

Scaling via Backprop & Adam

- Training a model with billions of parameters would be impossible without BP
- Stochastic Gradient Descent can be too slow on massive datasets, industry relies on sophisticated optimizers like Adam to safely navigate complex, high-dimensional loss landscapes

Takeaway

The fundamental rule remains the same whether you are *training a 3-layer network on a laptop* or *a trillion-parameter model across Google's TPUs*: you are approximate complex functions, and minimizing loss via the chain rule

Unsupervised Learning

Machine Learning Paradigm

Supervised: learn f from (x, y) → model learns mapping

Unsupervised: only x → discover structure, or latent features

- It discovers hidden patterns or latent structures in data
- When new data arrives, the model can represent it based on learned structure

Semi-supervised: hybrid of both worlds

- Uses small labeled set to guide large unlabeled pool.

Clustering with K-Means

What is Clustering? 👯

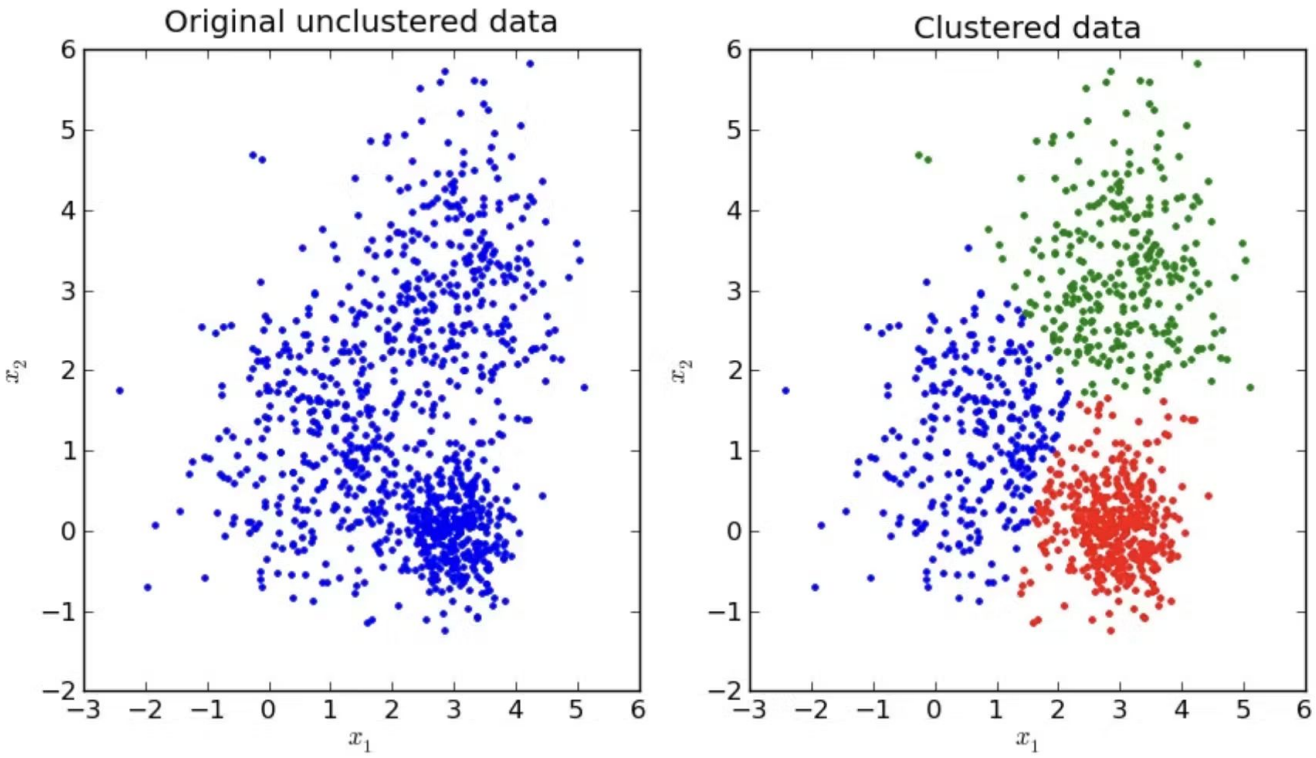
Goal: group similar data points into clusters based on similarity

Each cluster represents a natural “pattern” or “category”

Scenarios: segmenting customers, species, or document topics

🤔 No labels - the algorithm must decide grouping itself

K Clusters


Original unclustered data
Clustered data
x_1
x_2
−3 −2 −1 0 1 2 3 4 5 6
−2 −1 0 1 2 3 4 5 6

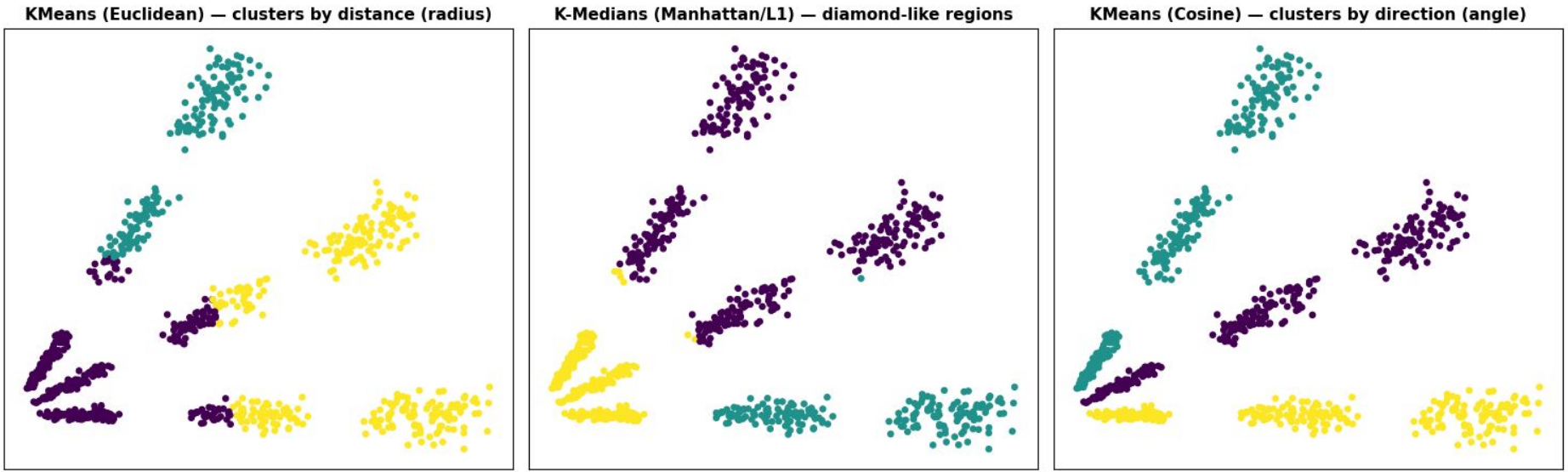
Similarity Metrics 📏

Similarity defines how close two samples are

Common distances:

- Euclidean → geometric closeness
- Manhattan → sum of absolute differences
- Cosine → angle between vectors (useful for text)

Comparisons



The K-Means Clustering Objective 🎯

Partition data into K clusters that minimize within-cluster distance

- Each cluster is represented by its centroid (mean vector)
- During optimization, each point is assigned to the nearest centroid, and the centroid is recomputed as the mean of all points in that cluster:

$$\min_{\{\mu_k\}_{k=1}^{K}} \sum_{i=1}^{N} \|x_i - \mu_{c_i}\|^2 \quad \text{where} \quad \mu_k = \frac{1}{|C_k|} \sum_{x_i \in C_k} x_i$$

Algorithm Overview ⚙️

Step 1 Initialize K random centroids

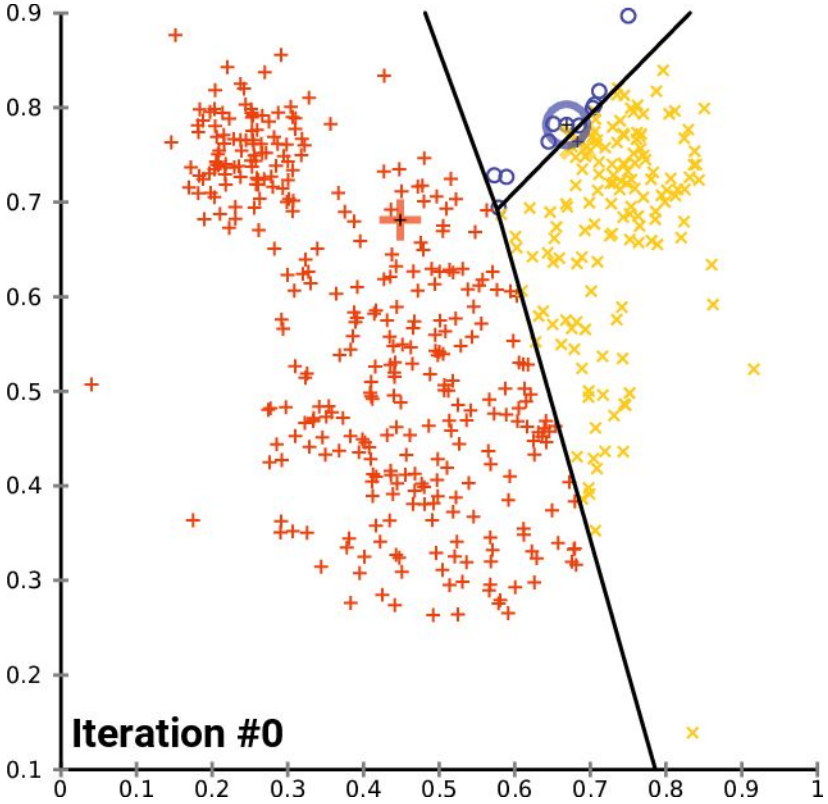
Step 2 Assign each point to its nearest centroid

Step 3 Re-calculate centroids as cluster means

Step 4 Repeat until assignments no longer change

Result → clusters + centroids

Algorithm Overview



Iterative Refinement 🔁

K-Means alternates between two simple steps:

- Assignment: label each point → nearest centroid
- Update: recalc centroid → mean of assigned points

Converges when centroids stop moving

Stops after a few iterations

K-means Strengths & Limitations ⚖️

✅ Simple and fast – scales well for large data

✅ Intuitive – easy to interpret centroids

❌ Sensitive to initialization and outliers

❌ Assumes spherical clusters, fails for complex shapes

❌ Requires manual K selection

Beyond K-Means 🌐

Hierarchical Clustering → tree-like structure

- Builds clusters step by step (divisive), visualized as a hierarchy

DBSCAN → density-based, detects arbitrary shapes

- Groups nearby points with high density and marks outliers as noise

Gaussian Mixture Models → probabilistic soft clustering

- Assigns each sample a probability of belonging to each cluster

Neural Network Fundamentals

Why Neural Networks? 💡

Many real-world problems are non-linear → linear models fail

- Relationships in data are curved, complex, and high-dimensional

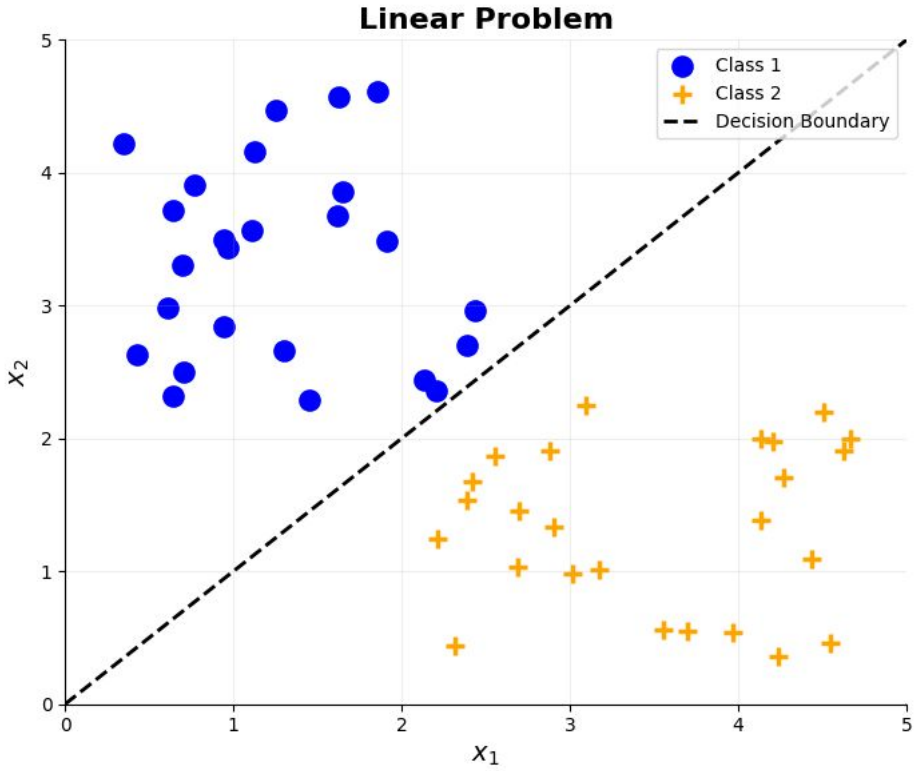
Neural networks can learn complex functions via composition

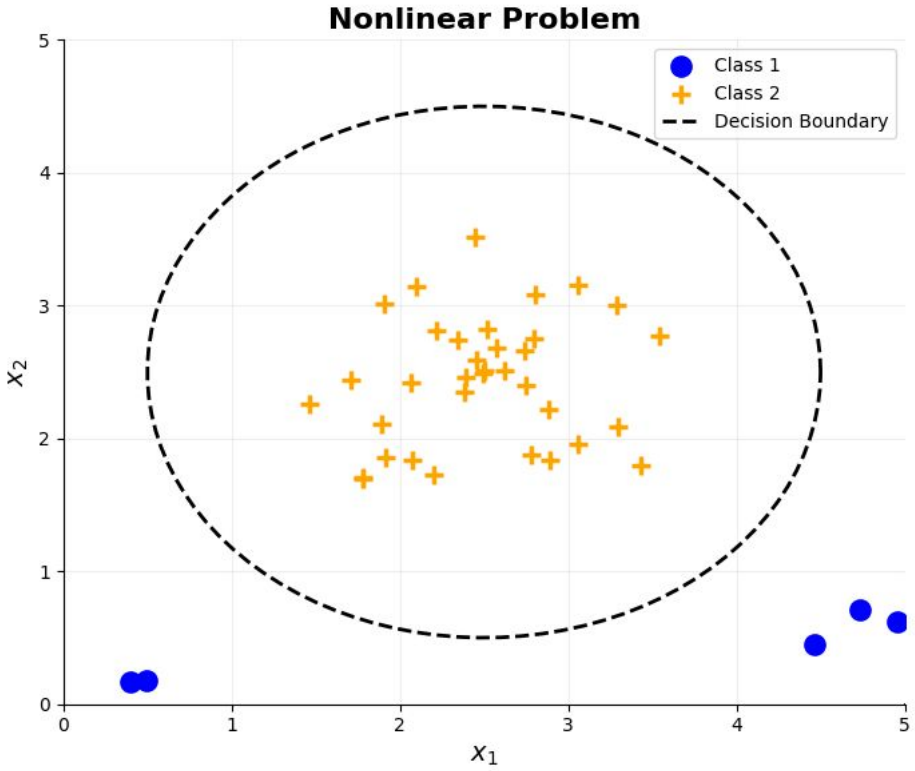
- By stacking neurons, we can approximate any function (Universal Approximation Theorem)

Inspired by the brain: neurons connect and adapt weights

- Each neuron adjusts its connection strength, gradually improving the generalizability

Linear vs Nonlinear





The Big Picture of Neural Networks 🧭

Start from linear models → limited expressiveness

Add nonlinear activations → learn complex functions

Stack layers → Multilayer Perceptron (MLP)

Train by forward + backward passes to minimize loss

Tune learning with optimization and regularization

Forms the foundation for CNNs, Transformers and ChatGPTs

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The Neuron ⚙️

Each neuron

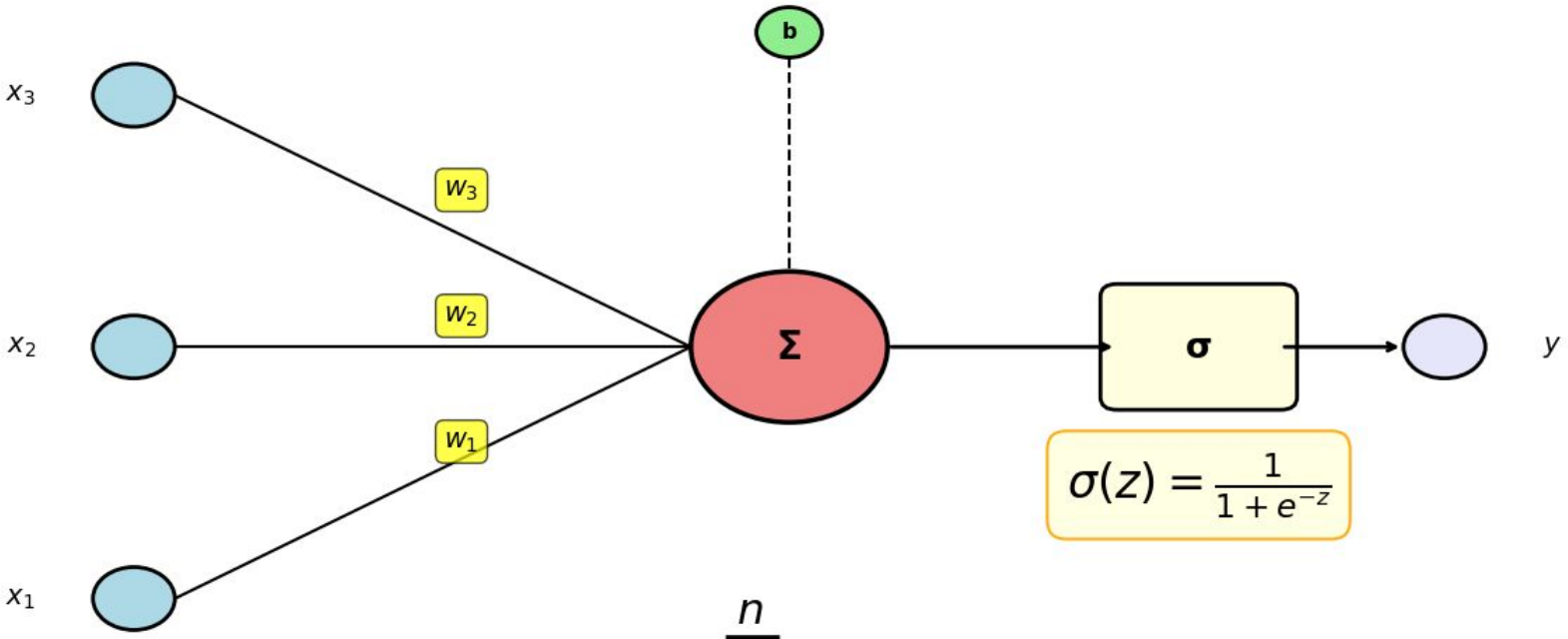
- A neuron receives various inputs and computes a weighted sum of them
- This sum is then passed through an activation function, which introduces non-linearity
- The result of this activation becomes the output of the neuron, serving as input for the next layer

The Neuron ⚙️

Nonlinear activations = flexible decision surfaces

- Without nonlinearity, the entire network would behave like a single linear model
- Nonlinear activations allow the network to learn complex, curved decision boundaries, making it capable of modeling intricate relationships in data

The Neuron 🧠



$$y = \sigma\left(\sum_{i=1}^{n} w_i x_i + b\right)$$

Activation Functions 🔌

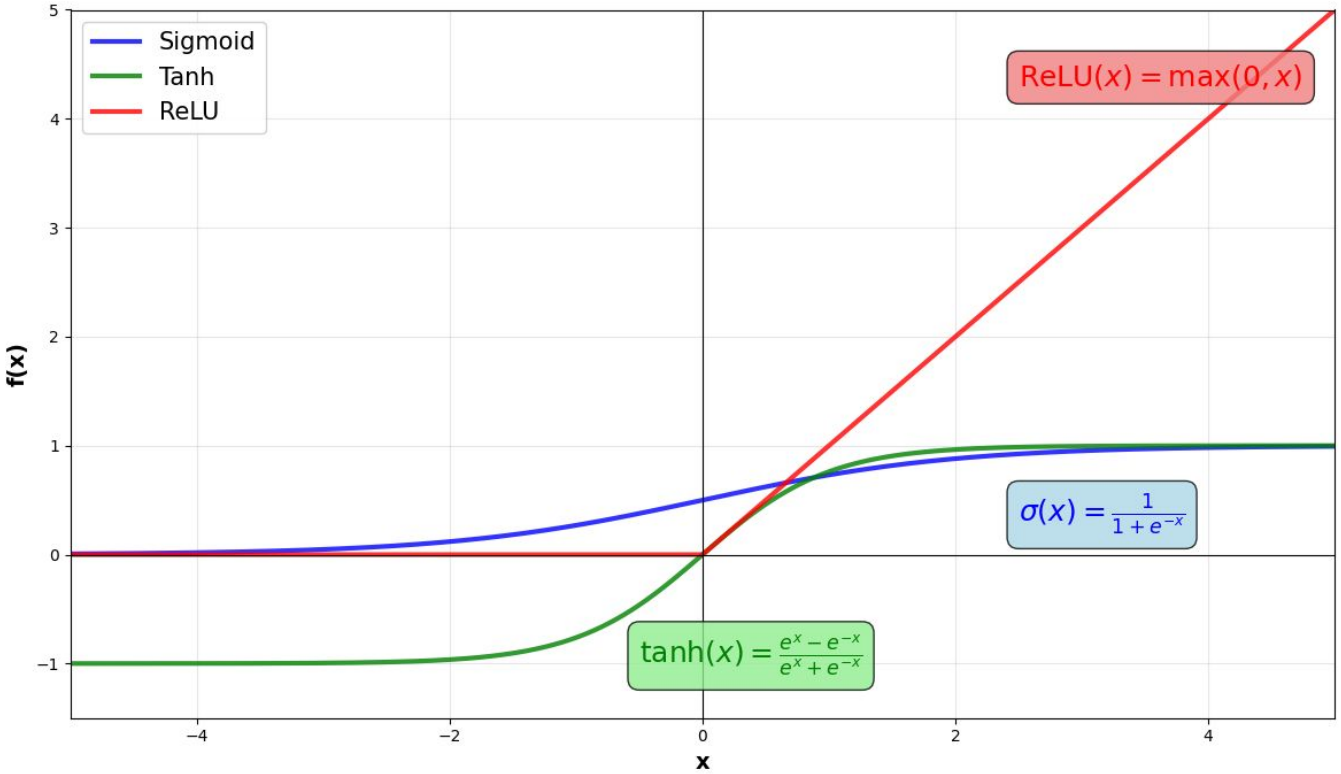
Sigmoid → squashes outputs (good for probabilities)

Tanh → zero-centered, faster convergence

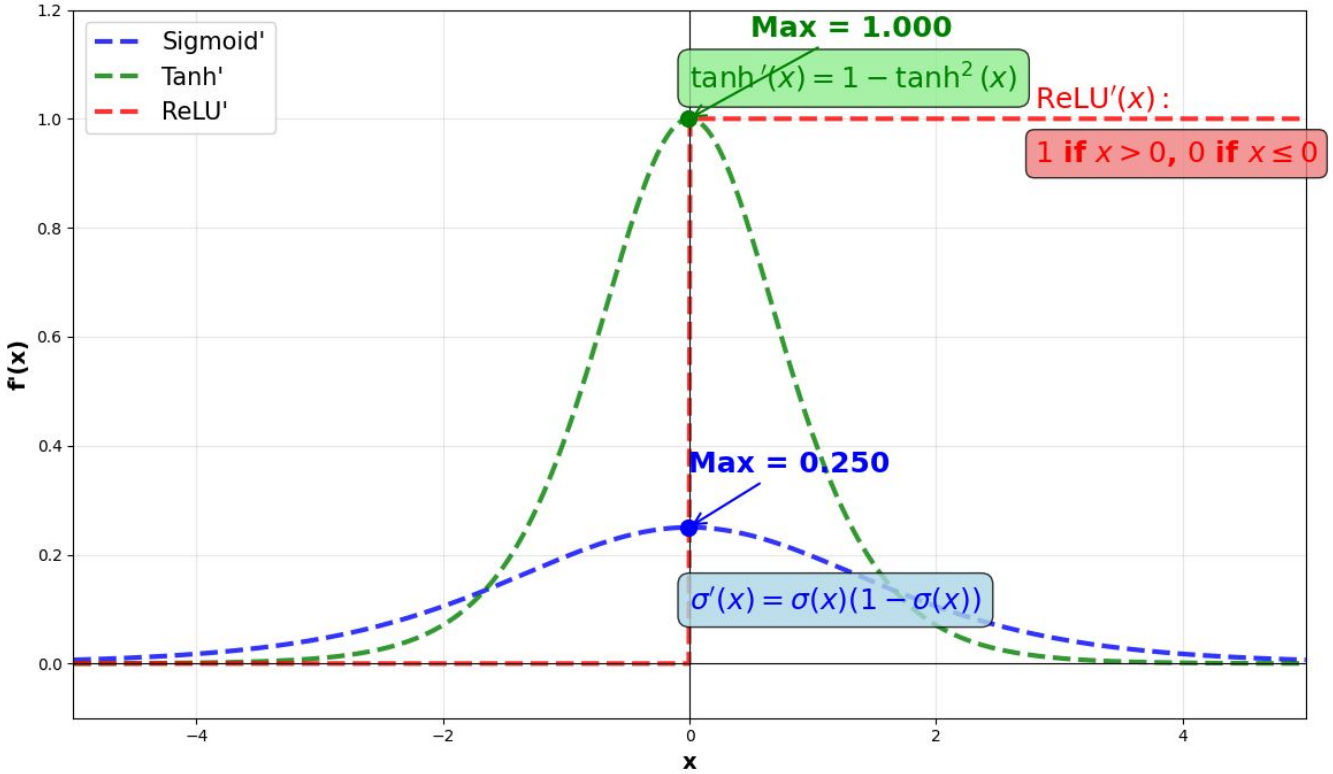
ReLU → sparse activation, efficient & dominant today

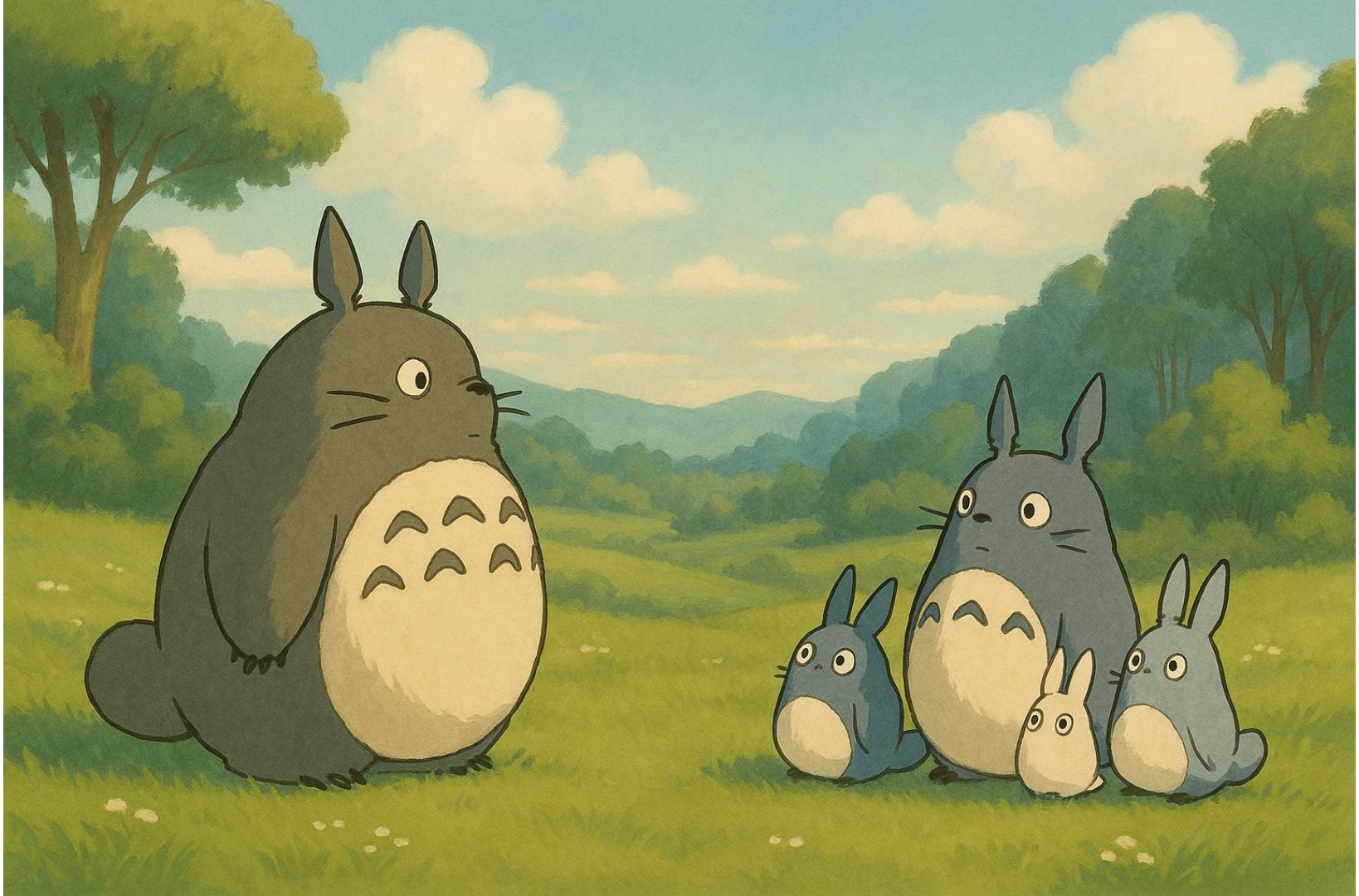
Modern variants: Leaky ReLU, GELU (used in Transformers)

Activation Functions Comparison



Derivatives Comparison





Multilayer Perceptron (MLP) 🏗️

Stack of Neurons

- A neural network is built by stacking many simple neurons (linear + activation) into layers

Hidden Layers → Feature Learning

- Each hidden layer transforms the data into a new representation - from raw input features to increasingly abstract ones

Multilayer Perceptron (Feed-forward) 🏗️

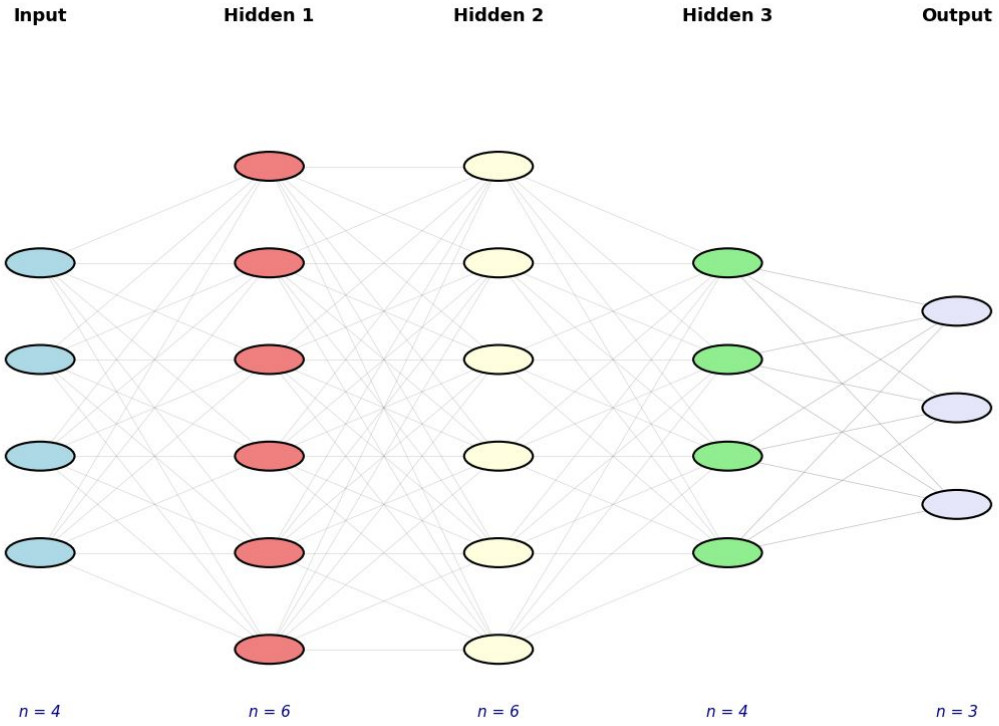
Output Layer → Task-Specific Prediction

- The final layer converts the learned representation into task-relevant outputs:
  - Regression → real values (e.g., price, temperature)
  - Classification → probabilities via Softmax (e.g., cat / dog / car)

🤔 As we go deeper, the network is not simply memorizing

- It automatically represents data using more abstract and meaningful features

Feedforward Architecture



Training Feedforward Network 🧠

Feedforward (left → right)

- Compute layer activations

Loss function (Distance between prediction and ground-truth)

- Map distance to a scalar error

Backprop (right → left)

- Use the computational graph and chain rule to compute gradients

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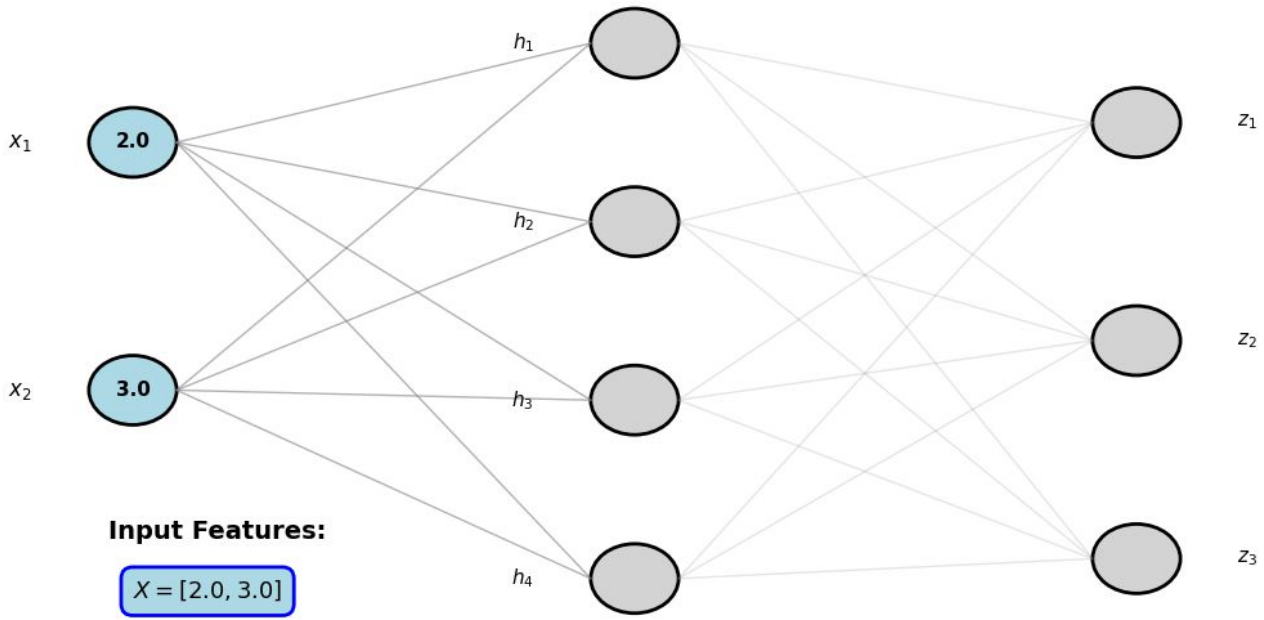
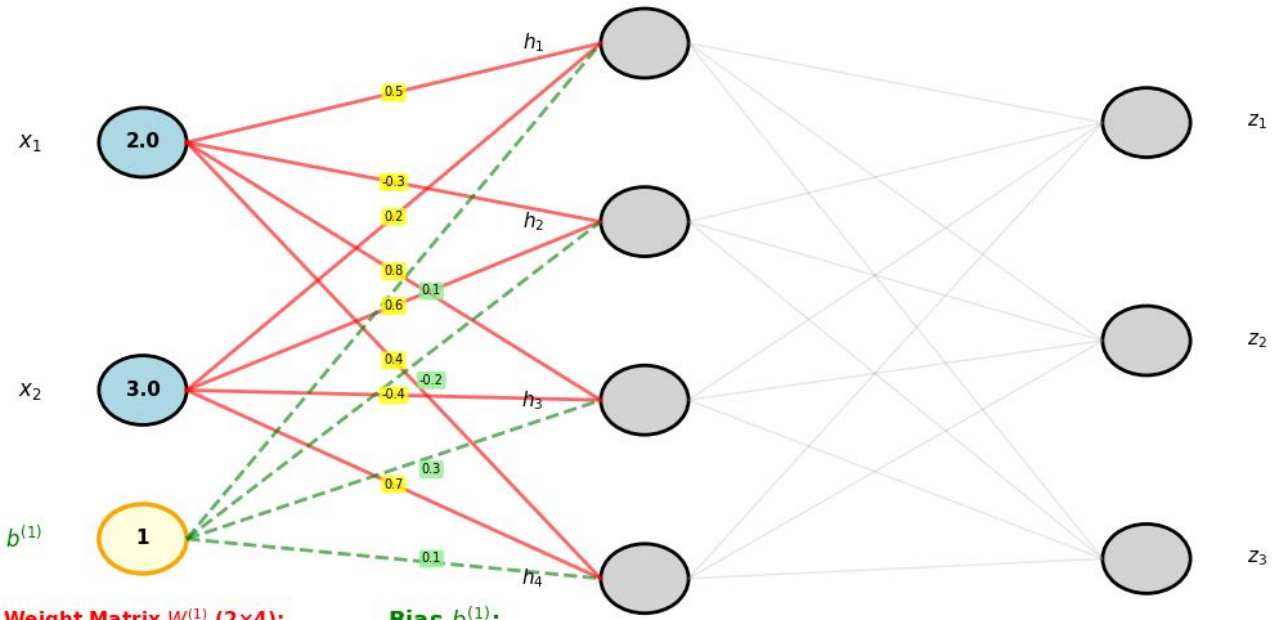

x_1
2.0
x_2
3.0
h_1
h_2
h_3
h_4
z_1
z_2
z_3
Input Features:
$X = [2.0, 3.0]$

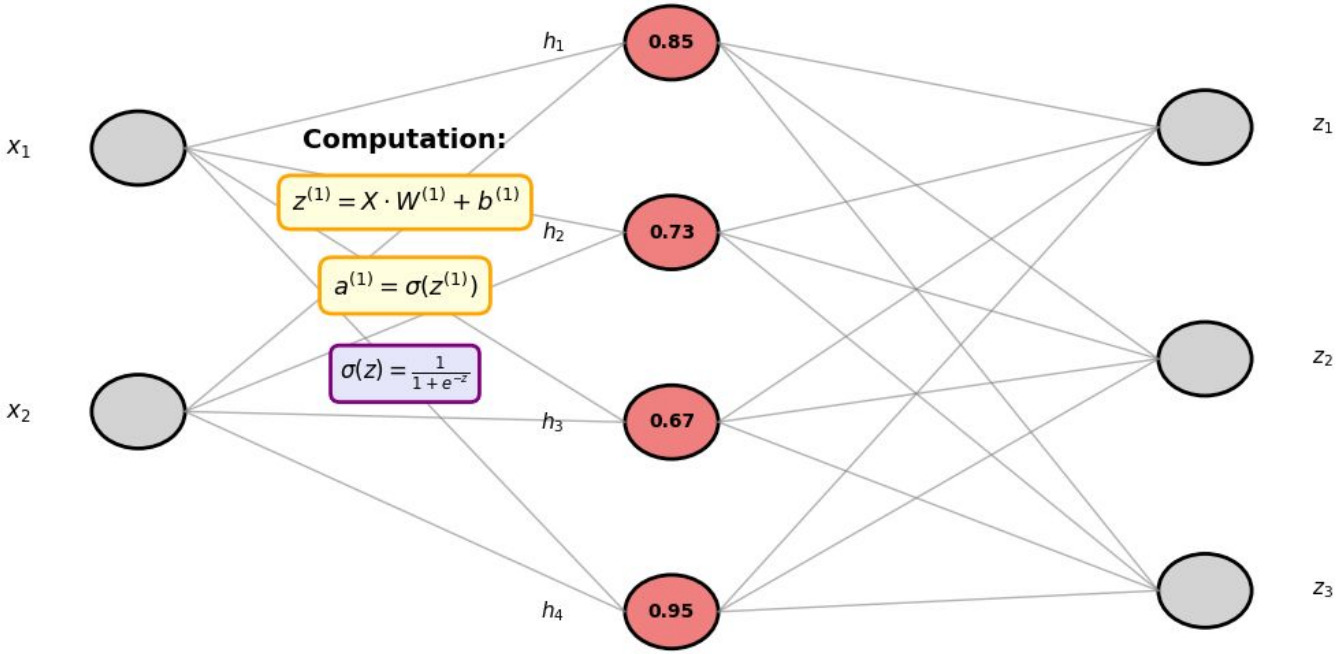


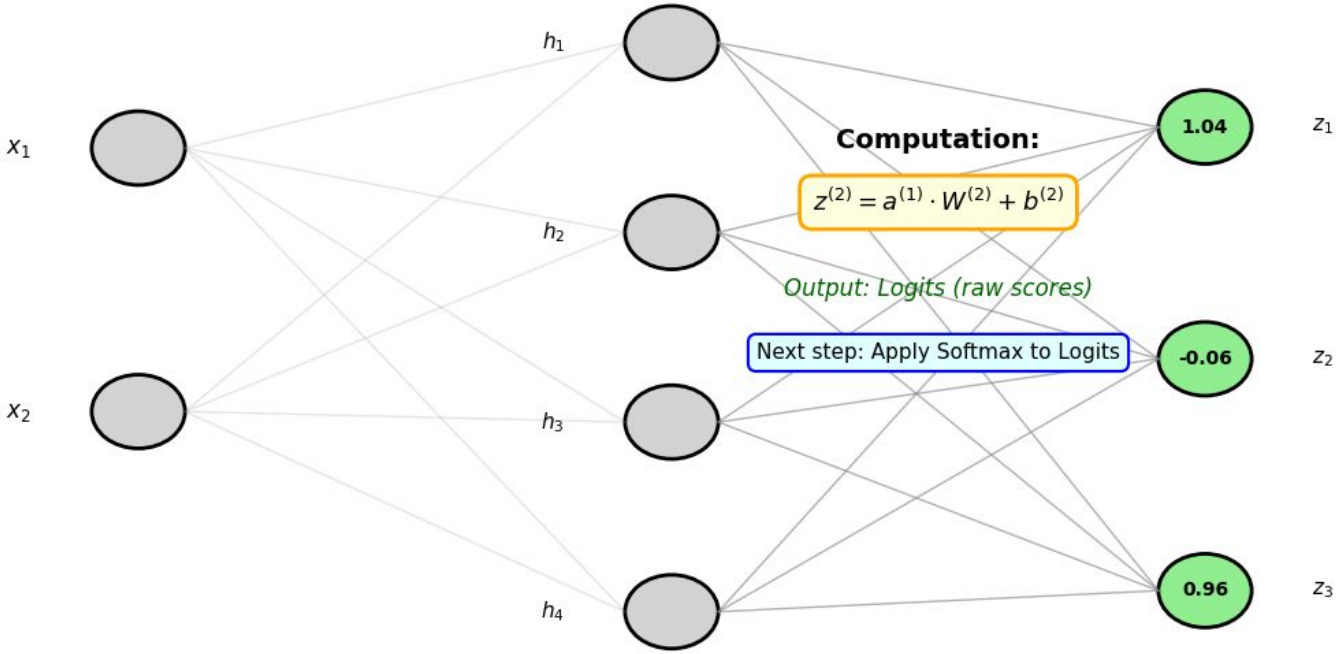
Weight Matrix $W^{(1)}$ (2×4):

$$\begin{bmatrix} 0.5 & -0.3 & 0.8 & 0.4 \\ 0.2 & 0.6 & -0.4 & 0.7 \end{bmatrix}$$

Bias $b^{(1)}$:

[0.1, -0.2, 0.3, 0.1]


x_1
x_2
Computation:
$z^{(1)} = X \cdot W^{(1)} + b^{(1)}$
$a^{(1)} = \sigma(z^{(1)})$
$\sigma(z) = \frac{1}{1+e^{-z}}$
h_1
h_2
h_3
h_4
0.85
0.73
0.67
0.95
z_1
z_2
z_3


x_1
x_2
h_1
h_2
h_3
h_4
Computation:
$z^{(2)} = a^{(1)} \cdot W^{(2)} + b^{(2)}$
Output: Logits (raw scores)
Next step: Apply Softmax to Logits
1.04
-0.06
0.96
z_1
z_2
z_3

Training Feedforward Network 🧠

Feedforward (left → right)

- Compute layer activations

Loss function (Distance between prediction and ground-truth)

- Map distance to a scalar error

Backprop (right → left)

- Use the computational graph and chain rule to compute gradients

Loss Functions 📉

Quantify how far predictions are from targets

Regression → MSE $$\mathrm{MSE} = \frac{1}{n}\sum_{i=1}^{n}(y_i - \hat{y}_i)^2$$

Classification → Cross-Entropy $$\mathrm{CrossEntropy} = -\sum_{i=1}^{n} y_i \log(\hat{y}_i)$$

Lower loss = better model fit on training data

Input: Logits

$z_1 = 1.04$

$z_2 = -0.06$

$z_3 = 0.96$

Softmax Function:

$$\text{Softmax}(z_i) = \frac{e^{z_i}}{\sum_{j=1}^{K} e^{z_j}}$$

Converts logits to probabilities

Output: Probabilities

$p_1 = 0.443$

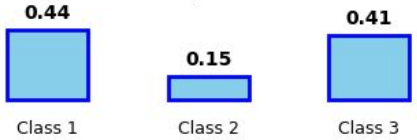
$p_2 = 0.148$

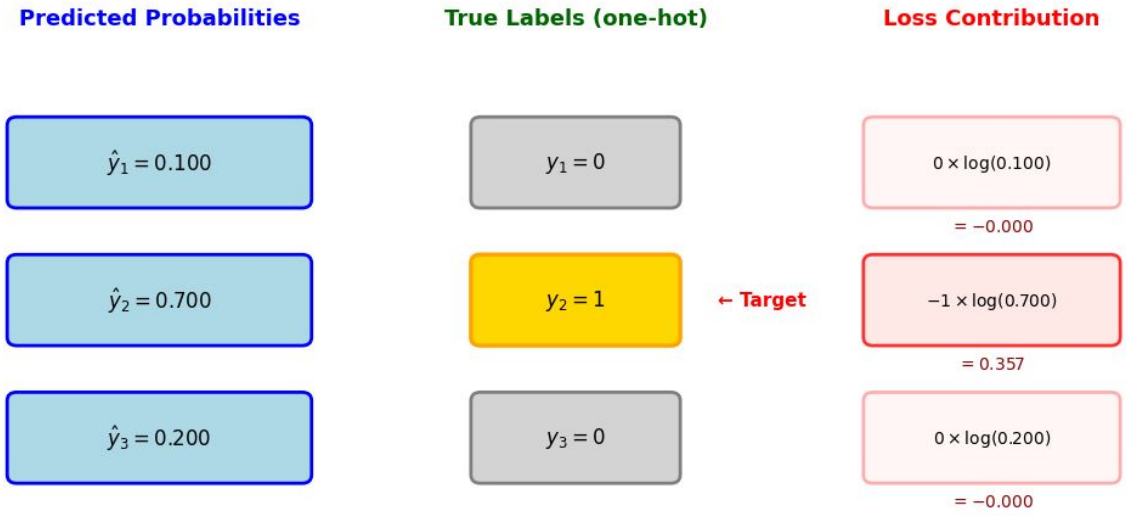
$p_3 = 0.409$

Properties:

- All probabilities: $0 \leq p_i \leq 1$
- Sum to 1: $\sum_i p_i = 1$

Probability Distribution:





$$L = -\sum_{i=1}^{K} y_i \log(\hat{y}_i)$$

Total Loss: $L = 0.3567$

Only the correct class contributes to loss

Training Feedforward Network 🧠

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Recap Gradient Descent

Update rule

$$\theta \leftarrow \theta - \eta \nabla_\theta L(\theta)$$

where \eta = learning rate/step size

Works for large datasets & online learning

Cornerstone of DL 🤖

Backpropagation 🔄

Efficient gradient computation using chain rule

- Backpropagation applies the chain rule to compute derivatives layer by layer

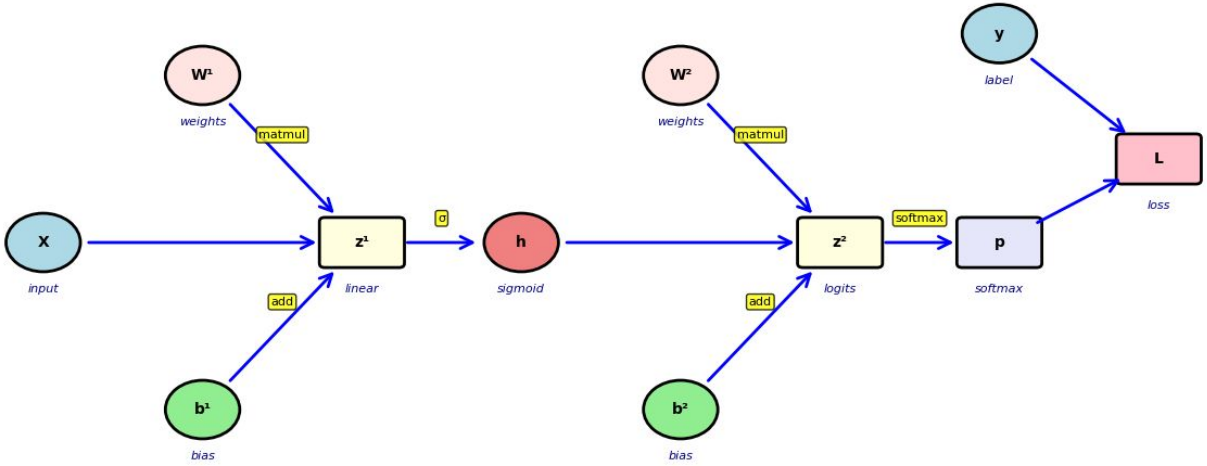
Gradients flow backward from output to input

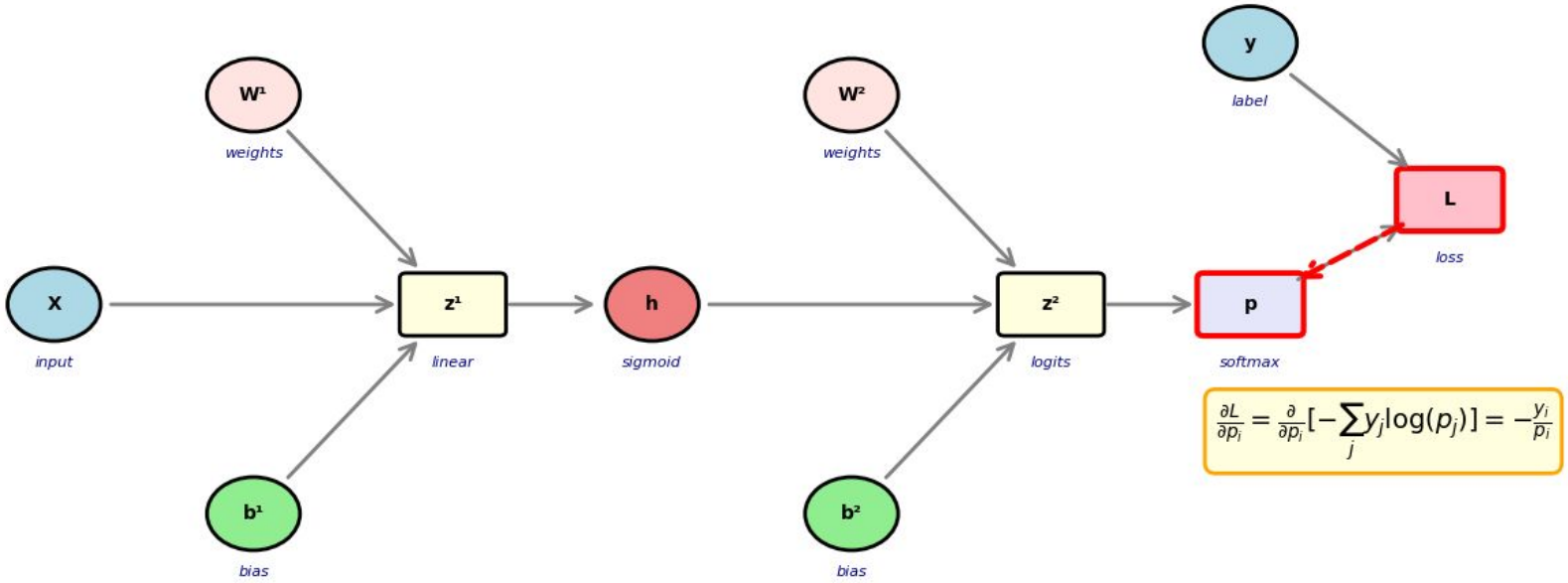
- Each layer receives gradients from the next layer to update its own weights

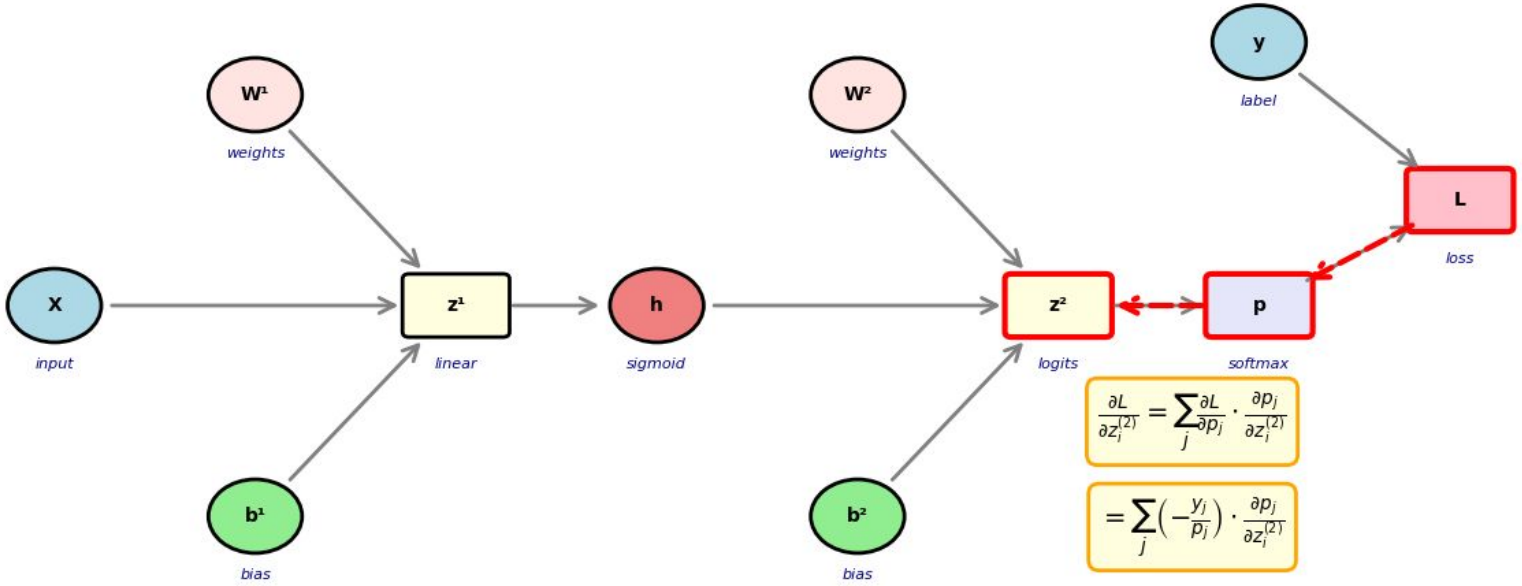
Enables training of deep networks with millions of weights

- Makes large-scale optimization computationally feasible
- Without backpropagation, deep learning would be impossible to train efficiently

Computational Graph




W¹
weights
W²
weights
y
label
L
loss
X
input
z¹
linear
h
sigmoid
z²
logits
p
softmax
b¹
bias
b²
bias
$\frac{\partial L}{\partial p_i} = \frac{\partial}{\partial p_i}[-\sum_j y_j \log(p_j)] = -\frac{y_i}{p_i}$



$$\frac{\partial L}{\partial z_i^{(2)}} = \sum_j \frac{\partial L}{\partial p_j} \cdot \frac{\partial p_j}{\partial z_i^{(2)}}$$

$$= \sum_j \left(-\frac{y_j}{p_j}\right) \cdot \frac{\partial p_j}{\partial z_i^{(2)}}$$

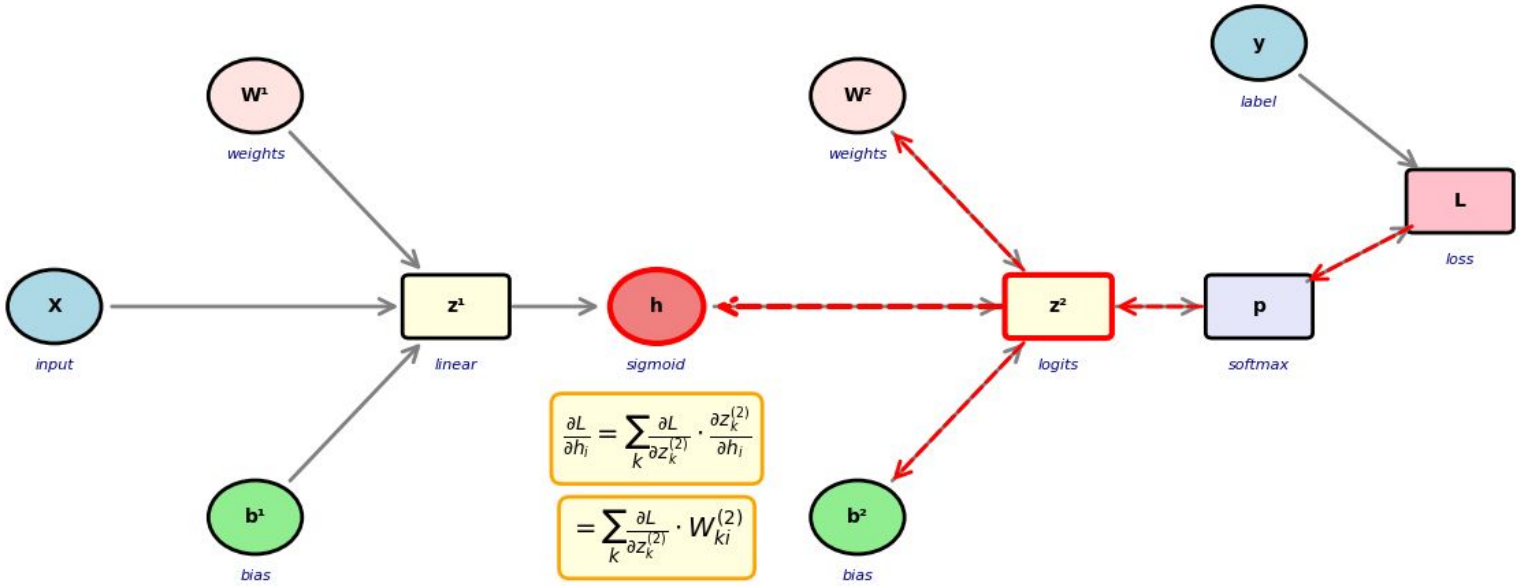
Softmax derivative:

if $i = j$: $\frac{\partial p_i}{\partial z_i} = p_i(1 - p_i)$

if $i \neq j$: $\frac{\partial p_j}{\partial z_i} = -p_i p_j$

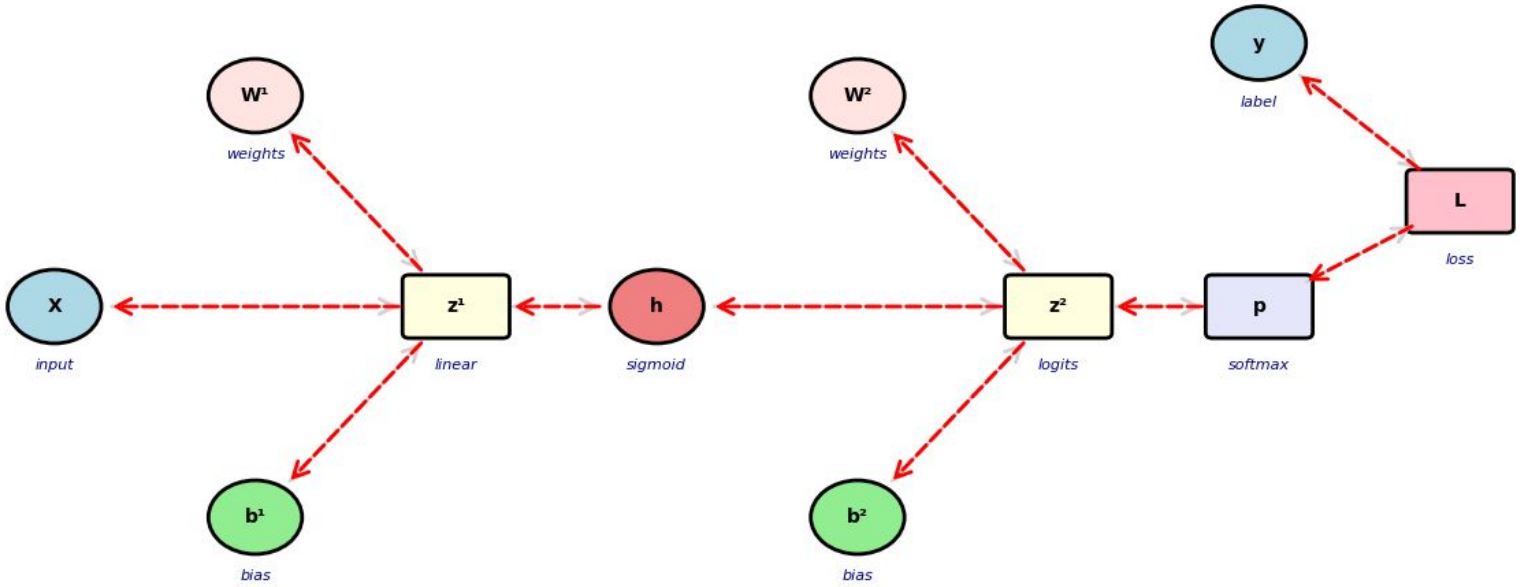
Final result:

$$\frac{\partial L}{\partial z_i^{(2)}} = p_i - y_i$$



$$\frac{\partial L}{\partial h_i} = \sum_k \frac{\partial L}{\partial z_k^{(2)}} \cdot \frac{\partial z_k^{(2)}}{\partial h_i}$$

$$= \sum_k \frac{\partial L}{\partial z_k^{(2)}} \cdot W_{ki}^{(2)}$$



Backpropagation Complete!

✓ Backward Pass: Compute all gradients (red dashed)

✓ Gradients computed via chain rule recursively

✓ Update: $\theta \leftarrow \theta - \eta \cdot \partial L/\partial \theta$ (Gradient Descent)

Optimizers ⚙️

Stochastic GD $\theta \leftarrow \theta - \eta \nabla_\theta L(\theta)$

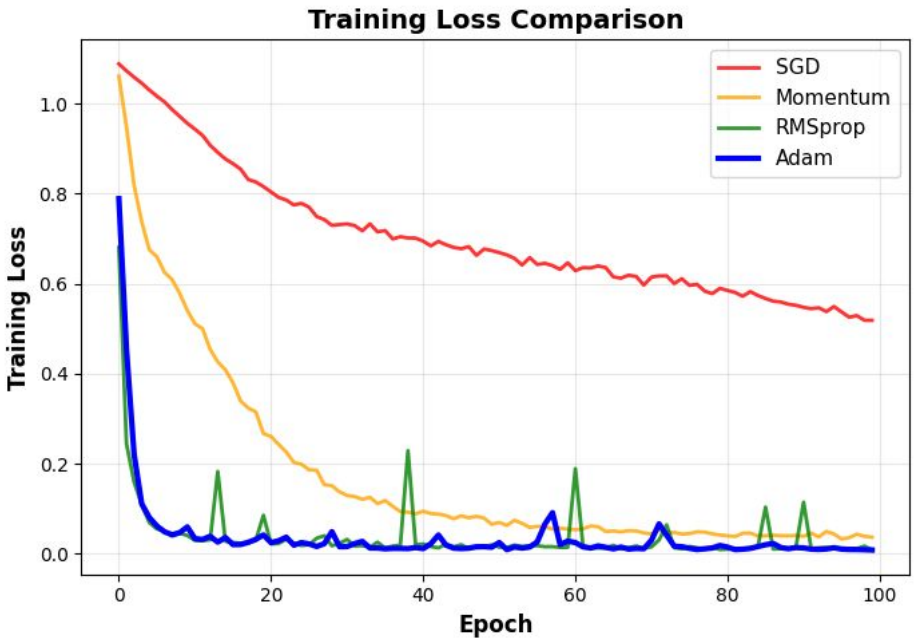
Too small → slow 🐢 too large → divergence 😳

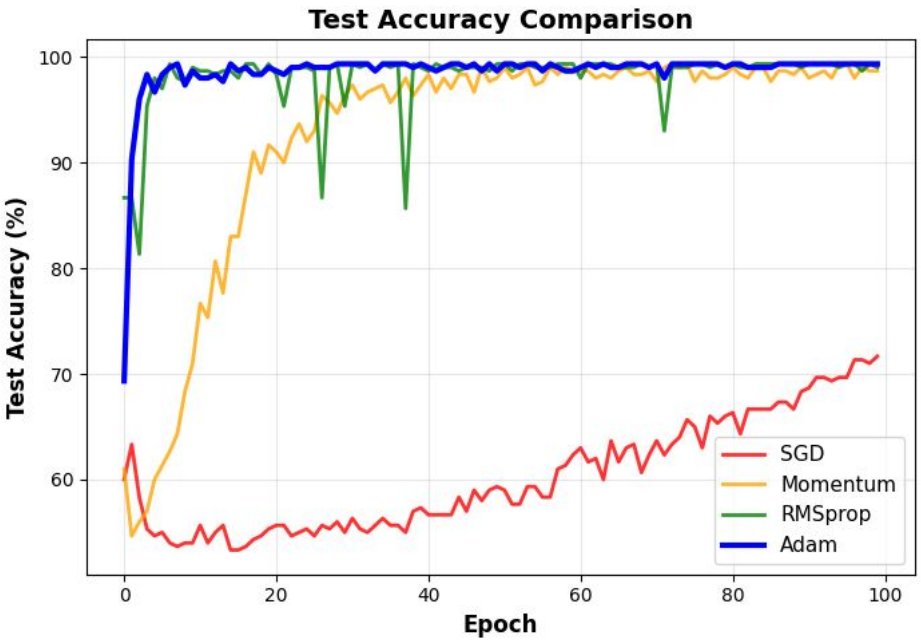
Momentum → Accelerates learning by smoothing gradients with past updates

RMSProp → Adapts learning rates based on recent gradient magnitudes

Adam → Momentum + RMSProp

Comparisons





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